Probabilistic Artificial Intelligence
Final Exam

Jan 30, 2021

Time limit:	120 minutes
Number of pages:	21
Total points:	78

You can use the back of the pages if you run out of space. Collaboration on the exam is strictly forbidden. Please write your answers with a *pen*.

Instructions: In this exam you will encounter the following question types.

- **Single choice questions.** There is only one correct answer out of several (typically, but not necessarily, 4). Said differently, no multiple choice question contains multiple true answers.
- **Groups of multiple True/False questions.** Each True/False question inside a group is evaluated separately. For example, if you answer 3 True/False correctly out of a group of 4, you get 75
- **Numerical answers. Answer using the decimal point system (not fractions) and round your answers up to two decimal places.** (The tolerance is +/- 0.005 around the correctly rounded value.) Only the final answer counts for the grade.

The notation $\mathcal{N}(\mu, \sigma^2)$ will designate a Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2. The acronym *i.i.d.* stands for *independently and identically distributed*.

Good luck!

(0 points) Please fill in your student ID and full name (LASTNAME, FIRSTNAME) in capital letters.

Please leave the table below empty.

Problem	**Maximum points**	**Obtained**
1.	7	
2.	17	
3.	4	
4.	20	
5.	6	
6.	15	
7.	9	
Total	78	

1. Kalman Filter (10 points)

Consider the following dynamical system describing a moving object

$$x_{t+1} = x_t + v_t, \quad v_{t+1} = av_t + \epsilon_t \,, \tag{D}$$

where $x_t \in \mathbb{R}$ denotes the position at time $t > 0$, $v_t \in \mathbb{R}$ denotes the velosity, $a > 0$ is an unknown damping parameter, and $\epsilon_t \in \mathbb{R}$ is additive noise with a time-independent and identically distributed (i.i.d.) Gaussian distribution $\mathcal{N}(0, \sigma_x^2)$ which models wind blowing in random directions. Consider also a noisy sensor measuring only the position x_t as

$$y_t = x_t + \eta_t, \tag{S}$$

where the i.i.d. noise η_t verifies $\eta_t \sim \mathcal{N}(0, \sigma_y^2)$.

(1 point) (i) Seeing the above systems (D)-(S) as a Kalman filter, what are the matrices F for the motion model and H for the sensor model, and their respective noise variance matrices Σ_x and Σ_y?

◯ $F = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $H = \begin{bmatrix} 1 & 1 \\ 0 & a \end{bmatrix}$, $\Sigma_x = \sigma_y^2$, $\Sigma_y = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_x^2 \end{bmatrix}$.

√ $F = \begin{bmatrix} 1 & 1 \\ 0 & a \end{bmatrix}$, $H = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $\Sigma_x = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_x^2 \end{bmatrix}$, $\Sigma_y = \sigma_y^2$.

◯ $F = \begin{bmatrix} 1 & 1 \\ 0 & a \end{bmatrix}$, $H = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $\Sigma_x = \begin{bmatrix} 0 & \sigma_x^2 \\ 0 & 0 \end{bmatrix}$, $\Sigma_y = \sigma_y^2$.

◯ $F = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $H = \begin{bmatrix} 1 & 1 \\ 0 & a \end{bmatrix}$, $\Sigma_x = \sigma_y^2$, $\Sigma_y = \begin{bmatrix} 0 & \sigma_x^2 \\ 0 & 0 \end{bmatrix}$.

(2 points) (ii) Assume that $\sigma_x = \sigma_y = 1$, $(x_0, v_0) \sim \mathcal{N}(\begin{bmatrix} 2 & 0 \end{bmatrix}^T, \sigma^2 I)$ with $\sigma = 1$, $a = 1$ and $y_1 = 5$. Knowing that the Kalman gain at time $t = 1$ is given by $K_1 = \frac{\sigma^2}{2\sigma^2 + \sigma_y^2} \begin{bmatrix} 2 & a \end{bmatrix}^T$, what is the Kalman update of the mean μ_1?

◯ $\mu_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ √ $\mu_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ ◯ $\mu_1 = \begin{bmatrix} 3.5 \\ 1.8 \end{bmatrix}$ ◯ $\mu_1 = \begin{bmatrix} 4.2 \\ 0.95 \end{bmatrix}$

Solution:

$$\begin{aligned}\mu_{t+1} &= F\mu_t + K_{t+1}(y_{t+1} - HF\mu_t) \\ &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \left(5 - \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \frac{1}{3}(5-2) \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}\end{aligned}$$

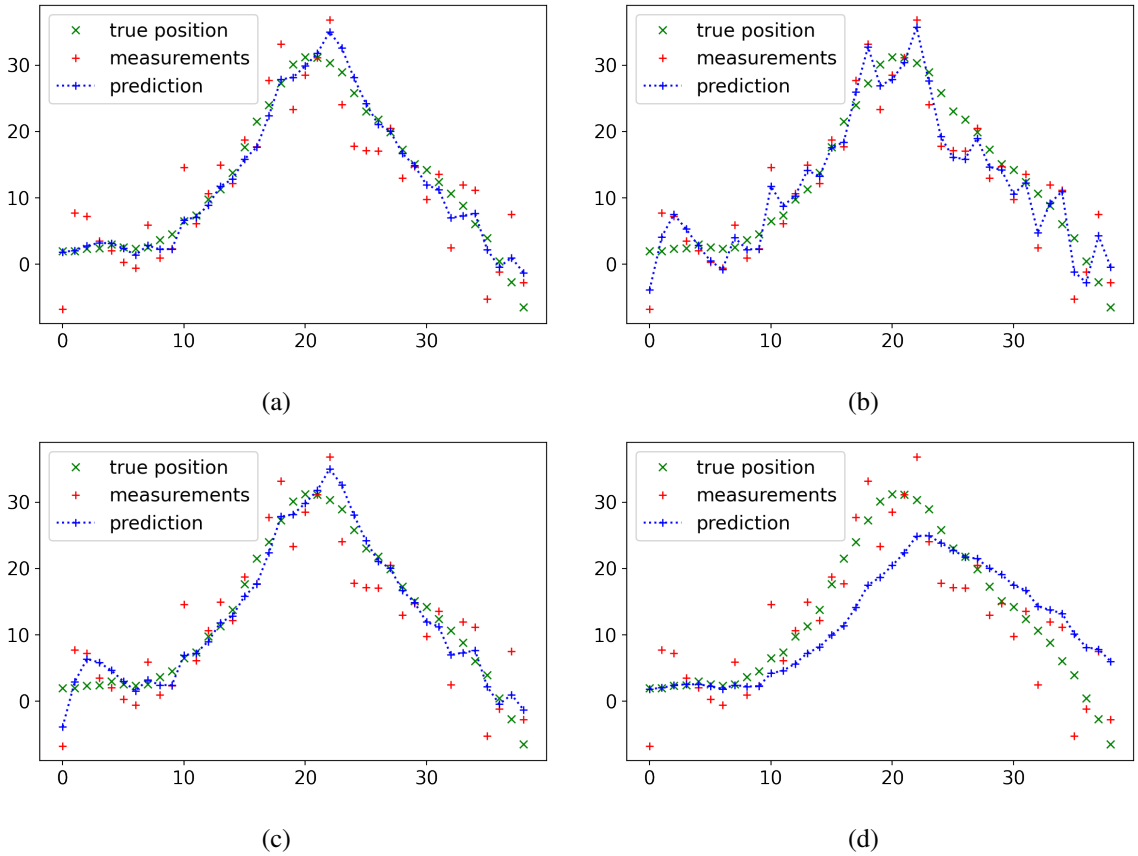


Figure 1: Positions x_t and measurements y_t for the system (D)-(S) as a function of time t, together with Kalman filtered values.

(4 points) (iii) Figure 1 shows data generated by the system (D)-(S) together with Kalman filtered positions. The true positions $(x_t)_t$ are plotted in green as a function of time t and used $a = 1$ and $\sigma_x = 1$. The measurements $(y_t)_t$ are red. The Kalman filtered values are blue and they all use $\sigma_x = 0$ and an

initial prior at $t = 0$ given by $(x_0, v_0) \sim \mathcal{N}(\begin{bmatrix} 2 & 0 \end{bmatrix}^T, \sigma^2 I_2)$. However, the Kalman filter uses different values of a, σ, and σ_y for every plot. Please assign each combination (a, σ, σ_y) to the correct plot.

1. $a = 1, \sigma = 10, \sigma_y = 10$ ◯ Fig.1a ◯ Fig.1b √ **Fig.1c** ◯ Fig.1d
2. $a = 1, \sigma = 1, \sigma_y = 1$ ◯ Fig.1a √ **Fig.1b** ◯ Fig.1c ◯ Fig.1d
3. $a = 1, \sigma = 1, \sigma_y = 10$ √ **Fig.1a** ◯ Fig.1b ◯ Fig.1c ◯ Fig.1d
4. $a = 0.5, \sigma = 1, \sigma_y = 10$ ◯ Fig.1a ◯ Fig.1b ◯ Fig.1c √ **Fig.1d**

Solution: Bigger values of σ_y lead to smoother filters. Hence Fig.1b corresponds to $\sigma_y = 1$ and all others to $\sigma_y = 10$. Smaller values of the prior variance parameter σ mean that the first filtered value will be closer to the prior mean position x_0. Hence Fig.1c corresponds to $\sigma = 10$ and the others to $\sigma = 1$. Finally, using $a = 0.5$ instead of the true value $a = 1$ introduces bias in the filter. Hence Fig.1d corresponds to $a = 0.5$ and all others to $a = 1$.

2. Gaussian Processes (6 points)

The Wiener process, also known as Brownian motion, is a Gaussian process $\mathcal{GP}(\mu_W, k_W)$ defined for $x \geq 0$, with mean function $\mu_W(x) = 0$ and covariance $k_W(x, y) = \min(x, y)$ for all $x, y \geq 0$.

(1 point) (i) Is there a point x for which the value of the Wiener process is deterministic (i.e., is fixed with probability 1) and, if yes, which one?

- ◯ $x = 0.5$
- √ $x = 0$
- ◯ $x = 1$
- ◯ No, there is none.

Solution: Yes, the point $x = 0$, because the variance there is 0.

(1 point) (ii) What is the expectation of the process in $x = \frac{1}{2}$?

Solution: The mean in $x = 0.5$ is $\mu_W(0.5) = 0$.

(2 points) (iii) Let $f \sim \mathcal{GP}(\mu_W, k_W)$ be a random draw from the Wiener process. What is the variance $\mathrm{Var}(f(x) - f(y))$ of $f(x) - f(y)$ for any $x, y \geq 0$?

- ◯ $\mathrm{Var}(f(x) - f(y)) = x - y$
- ◯ $\mathrm{Var}(f(x) - f(y)) = x - \min(x, y)$
- √ $\mathrm{Var}(f(x) - f(y)) = |x - y|$
- ◯ $\mathrm{Var}(f(x) - f(y)) = (x - y)^2$

Solution:

$$\begin{aligned}\mathrm{Var}(f(x) - f(y)) &= \mathrm{Var}(f(x)) + \mathrm{Var}(f(y)) - 2\mathrm{Cov}(f(x), f(y)) \\ &= \min(x, x) + \min(y, y) - 2\min(x, y) = |x - y|.\end{aligned}$$

(3 points) (iv) Assume now that you observe that $f(1) = 2$. What is the expectation of f in $x = 0.5$?

Solution: The posterior mean of a Wiener process is a linear interpolation between the datapoints (including the fixed point $f(0) = 0$), and a constant on the right of the last data point x. Therefore $E[f(1/2)] = f(1)/2 = 1$ and $E[f(1.5)] = 2$. This can also be found using the usual posterior mean formula for Gaussian process regression: $\mu'(x) = \mu(x) + k(x, 1)(k(1, 1))^{-1}2 = 2\min(x, 1)$.

(3 points) (v) Still assuming that $f(1) = 2$, what is the expectation of f in $x = 1.5$?

Solution: $f(1.5) = 2$.

(2 points) (vi) A Brownian bridge is a Wiener process for which one imposes that its random draws f satisfy $f(1) = 0$. Said differently, it is a Wiener process for which you observe that $f(1) = 0$. What is the kernel k_{BB} of the Brownian bridge for $0 \leq x, y \leq 1$?

- $\surd$ $k_{BB}(x, y) = \min(x, y) - xy$
- ◯ $k_{BB}(x, y) = \min(x, y)$
- ◯ $k_{BB}(x, y) = \min(x, y) + xy$
- ◯ $k_{BB}(x, y) = \min(x, y) - xy + 1$

Solution: The answer must satisfy $k_{BB}(1, 1) = 0$ (i.e., the variance at $x = 1$ must be 0), since the value of $f(1)$ is fixed. There is only one answer satisfying this criterium.

(5 points) (vii) You observe six datapoints depicted as red dots in the plots below (Figure 2). Assume that the datapoints (x, y) were generated from the model $y = f(x) + \epsilon$, where f comes from a Gaussian process with prior mean 0 and unknown kernel and ϵ is i.i.d. Gaussian noise. Now imagine that you apply Gaussian process regression using a prior $\mathcal{GP}(0, k)$ with mean 0 and that you test the following kernels.

1. Gaussian kernel: $k(x, x') = \exp(-\frac{(x-x')^2}{2.5^2})$.
2. Laplace kernel: $k(x, x') = \exp(-2|x - x'|)$.
3. Wiener kernel: $k(x, x') = .5 \min(x, x')$ for $x, x' \geq 0$.
4. Cosine kernel: $k(x, x') = \cos(|x - x'|)$.
5. Linear kernel: $k(x, x') = xx'$.

The obtained posterior mean (solid line) and posterior standard deviation (shaded area) are given in Figure 2 in no particular order, one plot for each kernel. Which plot corresponds to which kernel?

1. Gaussian kernel ◯ Fig.2a ◯ Fig.2b ◯ Fig.2c ◯ Fig.2d $\surd$ **Fig.2e**
2. Laplace kernel $\surd$ **Fig.2a** ◯ Fig.2b ◯ Fig.2c ◯ Fig.2d ◯ Fig.2e
3. Wiener kernel ◯ Fig.2a $\surd$ **Fig.2b** ◯ Fig.2c ◯ Fig.2d ◯ Fig.2e
4. Cosine kernel ◯ Fig.2a ◯ Fig.2b ◯ Fig.2c $\surd$ **Fig.2d** ◯ Fig.2e
5. Linear kernel ◯ Fig.2a ◯ Fig.2b $\surd$ **Fig.2c** ◯ Fig.2d ◯ Fig.2e

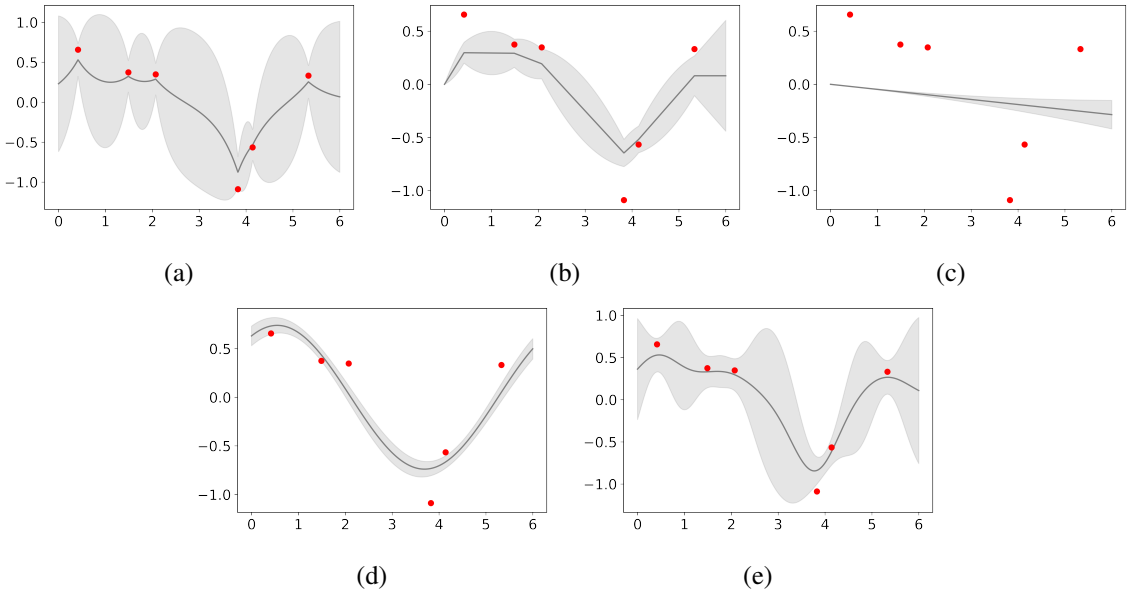


Figure 2: Posterior means (solid line) and standard deviations (shaded area) computed with GP regression using various kernels k (a different kernel for each plot). The data (red dots) is the same for every plot.

Solution: 1e, 2a, 3b, 4d, 5c. Both the Laplace and Wiener process will yield non-differentiable posterior means. But the Wiener process is restricted to 0 in 0, which helps discriminating between both. Moreover, the Laplace kernel is translation-invariant, which implies that the mean function will converge back to the prior mean (0) when moving away from data. The linear kernel can only produce linear mean functions, and the cosine kernel produces sine mean functions. And the Gaussian kernel produces very smooth, non-periodic mean functions.

3. Fitting a model

You are given some data $\mathcal{D} = \{X_i\}_{i=1}^{n=20}$ from an unknown discrete distribution supported on the non-negative integers $\mathbb{N}_0$, i.e., $X_i \overset{\text{i.i.d.}}{\sim} P$ with $X_i \in \mathbb{N}_0$ for all $i \in \{1, \ldots, 20\}$. The data is displayed in Table 1.

X_i	0	1	2	3	4	5
Frequency	5	5	2	0	4	4

Table 1: Observations and respective frequencies in the data $\mathcal{D}$. For example, you observed $X_i = 0$ five times and $X_i = 4$ four times and you did not observe $X_i = 3$.

You decide to approximate the unknown distribution P using a Poisson distribution $Q(\lambda)$ with parameter λ. Recall the probability mass function of the Poisson distribution for $k \in \mathbb{N}_0$:

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{if } X \sim Q(\lambda) \tag{1}$$

For this purpose, you would like to choose λ to minimize the forward KL divergence between the empirical distribution $\hat{P}$ associated with your data $\mathcal{D}$ and $Q(\lambda)$. This is, you want to compute

$$\lambda^* = \arg\min_\lambda \mathrm{KL}(\hat{P} \,\|\, Q) \tag{2}$$

(2 points) (i) Which of the following expression is equal to $\mathrm{KL}(\hat{P} \,\|\, Q)$, up to an additive and a positive multiplicative constant that are independent of λ?

- $\surd$ $20\lambda - 45 \log \lambda$
- ◯ $-20\lambda + 45 \log \lambda$
- ◯ $25\lambda - 45 \log \lambda$
- ◯ $20\lambda - 20 \log \lambda$

Solution: The definition of the KL-divergence together with the table give:

$$\begin{aligned}
\mathrm{KL}(\hat{P} \parallel Q) &= \sum_{i=0}^{5} \hat{P}(k=i) \log \frac{\hat{P}(k=i)}{Q(k=i)} \\
&= \frac{5}{20} \log \frac{5/20}{\lambda^0 e^{-\lambda}/0!} + \frac{5}{20} \log \frac{5/20}{\lambda^1 e^{-\lambda}/1!} + \frac{2}{20} \log \frac{2/20}{\lambda^2 e^{-\lambda}/2!} \\
&\quad + \frac{4}{20} \log \frac{4/20}{\lambda^4 e^{-\lambda}/3!} + \frac{4}{20} \log \frac{4/20}{\lambda^5 e^{-\lambda}/4!} \\
&= -\frac{1}{20}(5+5+2+4+4) \log e^{-\lambda} \\
&\quad - \frac{1}{20}(5 \cdot 0 + 5 \cdot 1 + 2 \cdot 2 + 4 \cdot 4 + 4 \cdot 5) \log \lambda + C \\
&= \frac{1}{20}(20\lambda - 45 \log \lambda) + C
\end{aligned}$$

where C is independent of λ.

(2 points) (ii) What is the value of λ^*?

Solution: (The correct answer is 2.25, but answers $1.$ and 1.8 should also get full points, since they would be consistent with answers $20\lambda - 20 \log \lambda$ and $25\lambda - 45 \log \lambda$ from the previous question.)

By differentiating the objective (see previous question) and searching its zeros, we find that $\lambda^* = 2.25$. (More generally, it can be shown that $\lambda^* = \frac{1}{n} \sum_i X_i$, the empirical mean of the data.)

4. Approximate inference methods and Bayesian neural networks

(4 points) (i) Which of the following statements about Bayesian logistic regression with a spherical Gaussian prior centered at zero, i.e., $\mathcal{N}\left(0, \sigma^2 I\right)$, on the regression parameters are true?

- The posterior distribution can be computed in closed form. ◯ True √ **False**
- The posterior log-density is convex. √ **True** ◯ False
- As σ increases, the mode of the posterior approaches the solution of standard (i.e., non-Bayesian) logistic regression. √ **True** ◯ False
- Variational inference using the multivariate Gaussian family with a full covariance matrix will recover the posterior distribution exactly. ◯ True √ **False**

Solution: The posterior distribution cannot be computed in closed form, but the log-density is convex (logistic regression is convex, regularizer is convex). As the prior becomes uninformative, MAP converges to the standard solution. No, the posterior is not Gaussian, so variational inference will not recover it exactly.

(4 points) (ii) You would like to approximate a target distribution $P(X = x) \propto \frac{1}{1+x^2}$ with $x \in \mathbb{R}$ by using the Metropolis-Hastings (MH) algorithm. Which of the following MH schemes are valid (i.e., the stationary distribution of the induced Markov chain is the target distribution)?

- Use proposal $R(x'|x) \sim \mathcal{N}(x, 1)$ and accept with probability $\alpha = \min\left\{1, \frac{(1+x^2)}{(1+(x')^2)}\right\}$. √ **True** ◯ False
- Use proposal $R(x'|x) \sim \mathcal{N}(x, 0.5)$ and accept with probability $\alpha = \min\left\{1, \frac{(1+(x')^2)}{(1+x^2)}\right\}$. ◯ True √ **False**
- Use proposal $R(x'|x) \sim \mathcal{N}(0, 1)$ and accept with probability $\alpha = \min\left\{1, \frac{(1+x^2)}{(1+(x')^2)}\right\}$. ◯ True √ **False**
- Use proposal $R(x'|x) \sim \mathcal{N}(0, 0.5)$ and accept with probability $\alpha = \min\left\{1, \frac{(1+x^2)}{(1+(x')^2)} \exp\left(x^2 - (x')^2\right)\right\}$. ◯ True √ **False**

Solution: The first option is standard textbook. The second option has the fraction, the wrong way round. The third option is missing the proposal ratio and the fourth option has the minus inside the exponent the wrong way round.

(4 points) (iii) Which of the following statements about Bayesian Neural Networks (BNNs) are true?

- BNNs are neural networks that parameterize a likelihood function and specify a prior distribution over their weights. √ **True** ◯ False
- If both the likelihood and the prior are Gaussian, MAP inference of the weights of a BNN can be performed in closed form. ◯ True √ **False**

- Training and evaluating an ordinary neural network with dropout can be interpreted as performing approximate inference on a BNN. √ **True** ◯ False
- Epistemic uncertainty in BNNs can be reduced by observing more data. √ **True** ◯ False

Solution: BNNs are neural networks that parameterize a likelihood function and specify a prior distribution over their weights (from the slides). MAP inference of the weights of a BNN cannot be performed in closed form, even if both prior and likelihood are Gaussian. Note that in general, the objective will be non-convex in the weights. Dropout has this variational interpretation. By definition, the epistemic uncertainty is the part of uncertainty that is due to not having an infinite amount of data. It therefore decreases with the amount of data.

(4 points) (iv) Which of the following methods can be used for approximate inference in Bayesian Neural Networks (BNNs)?

- Stochastic Gradient Langevin dynamics √ **True** ◯ False
- Monte Carlo Dropout √ **True** ◯ False
- Probabilistic Ensembles √ **True** ◯ False
- Black-box stochastic variational inference. √ **True** ◯ False

Solution: All methods are valid. See lecture slides.

(4 points) (v) Which of the following statements about Gaussian process (GP) regression are true?

- Gaussian process regression (with a Gaussian likelihood) admit predictions in closed-form. √ **True** ◯ False
- Only the posterior mean, but not the posterior covariance depends on the choice of hyperparameters of the covariance function. ◯ True √ **False**
- Fully independent training conditional (FITC) can be used to reduce the computational complexity (as a function of the number of data points) of GP methods at the expense of getting approximate results. √ **True** ◯ False
- Random Fourier features can be used to reduce the computational complexity (as a function of the number of data points) of GP methods at the expense of getting approximate results. √ **True** ◯ False

Solution: Gaussian process methods with a Gaussian likelihood admit predictions in closed-form. The posterior covariance depends on the kernel hyperparameters. Random Fourier Features can be used to reduce the computational complexity from $\mathcal{O}(n^3)$ to $\mathcal{O}(nm^2 + m^3)$. FITC is cubic in the number of inducing points but linear in the number of data points.

5. Bayesian optimization and acquisition functions (9 points)

In a typical Bayesian optimization problem, one wants to find the maximum of a function $f : x \mapsto f(x)$ with as little queries to f as possible. To do so, we assume that f is a draw from a Gaussian process (GP) $\mathcal{GP}(0, k)$ with known covariance function (a.k.a. kernel) k, from which we get t noisy observations $\{(x_i, y_i)\}_{i=1}^{t}$ that satisfy $y_i = f(x_i) + \epsilon_i$ where $\epsilon_1, \ldots, \epsilon_t$ is i.i.d. Gaussian Noise. We denote by $\mu_t(x)$ and $\sigma_t(x)$ the posterior mean and the posterior standard deviation at x of f after observing t points (x_i, y_i). To choose the next point x_{t+1} to query, we will use a so-called *acquisition function* a (our notation omits the dependence on t), and choose the point $x_{t+1} = \text{argmax}_x\, a(x)$ that maximizes this function. Typical acquisition functions are:

1. *Probability of improvement (PI)* a_{PI}, where $a_{PI}(x)$ represents the probability (w.r.t. our current posterior belief over f) that $f(x) \geq \mu_*$ at point x. Here, μ_* designates the maximum of the posterior mean at the queried points, i.e., $\mu_* := \max_{i=1,\ldots,t} \mu_t(x_i)$.
2. *Expected improvement (EI)* a_{EI}, where $a_{EI}(x)$ represents the expectation of the so-called *improvement* $u(f(x)) := \max(0, f(x) - \mu_*)$.
3. *Upper confidence bound (UCB)* $a_{UCB}(x) := \mu_t(x) + \beta\sigma_t(x)$, where β is an adjustable positive parameter.

(4 points) (i) Each of the 4 plots of Figure 3 show one of the following acquisition functions (solid line), in arbitrary order: PI, EI, UCB with $\beta = 0.5$, and UCB with $\beta = 2$. They were computed using the same GP posterior distribution (see posterior mean and standard deviations in gray as dotted line and shaded area respectively), using GP regression on the red dot datapoints (x, y) with prior mean 0 and the same prior Gaussian covariance function.

Please assign each of the plots to its corresponding acquisition function.

(Note that the left y-axis is for the acquisition function while the right one is for the datapoints and the GP posterior. Both axes have different scales.)

- Figure 3a ◯ PI ◯ EI √ **UCB with** $\beta = 0.5$ ◯ UCB with $\beta = 2$.
- Figure 3b ◯ PI √ **EI** ◯ UCB with $\beta = 0.5$ ◯ UCB with $\beta = 2$.
- Figure 3c √ **PI** ◯ EI ◯ UCB with $\beta = 0.5$ ◯ UCB with $\beta = 2$.
- Figure 3d ◯ PI ◯ EI ◯ UCB with $\beta = 0.5$ √ **UCB with** $\beta = 2$.

Solution: The UCB curves must pass through the datapoints (since, at the datapoints, $\sigma = 0$), which applies only to Figs.3a and 3d. (Beware, the y-scale is different for the acquisition function and the GP mean and std.). Between the two, Fig.3d clearly has larger variations, meaning it has a larger value of β. For Figs.3b and 3c, there is only EI and PI left, and since we expect that the probability of improvement should be highest near the highest observations, while expected improvement could also be high in regions with large standard deviation, we see that Fig.3b and 3c must correspond to EI and PI respectively.

(2 points) (ii) For each plot in Figure 3, i.e., each acquisition function, which point – (A), (B), (C) or (D) – will be queried next?

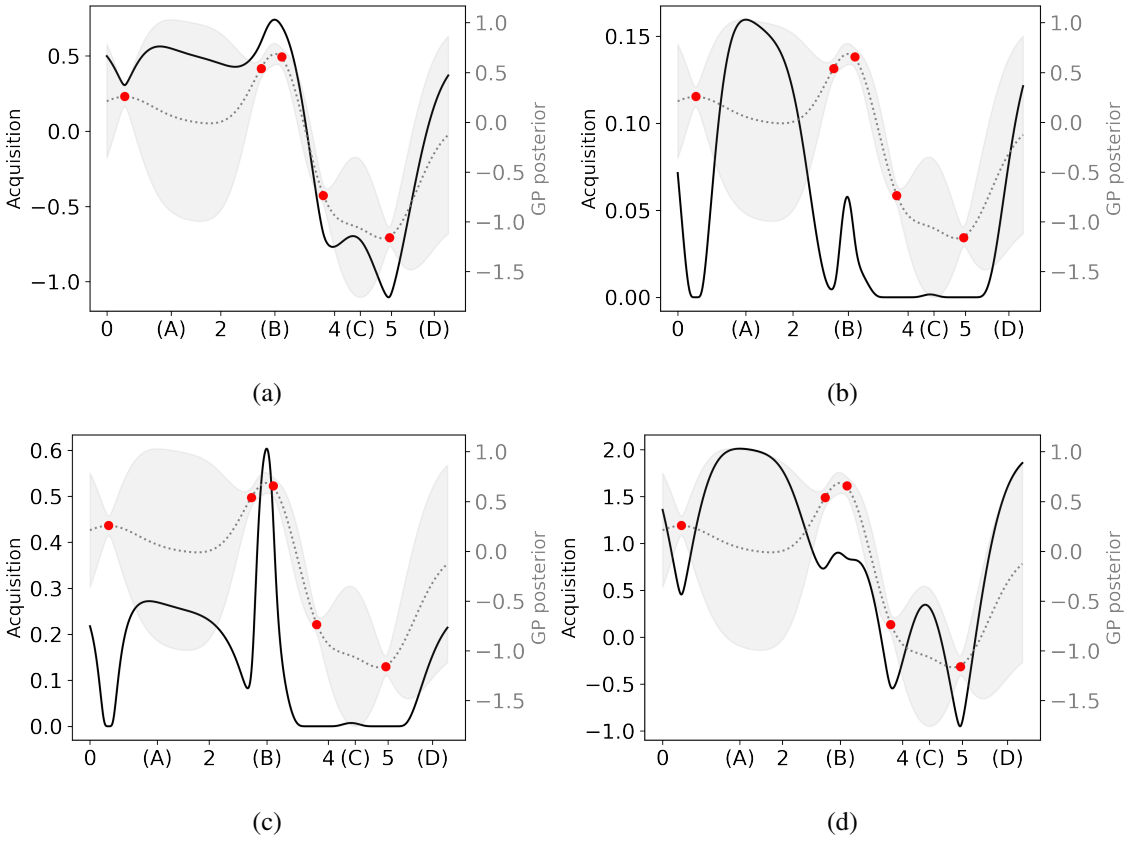


Figure 3: Different acquisition functions (one per plot) using the GP posterior distribution shown in gray and the red dot datapoins. (Beware: the y-scales are different for the acquisition function and the GP posterior.)

(Note that the left y-axis is for the acquisition function while the right one is for the datapoints and the GP posterior. Both axes have different scales.)

- Figure 3a: ◯ (A) √ **(B)** ◯ (C) ◯ (D)
- Figure 3b √ **(A)** ◯ (B) ◯ (C) ◯ (D)
- Figure 3c ◯ (A) √ **(B)** ◯ (C) ◯ (D)
- Figure 3d √ **(A)** ◯ (B) ◯ (C) ◯ (D)

Solution: The next point x to query is given by the maximum of the acquisition function.

6. Winning the next race (15 points)

In a legendary race, the Rabbit fell asleep and lost the race against the Turtle. Your goal is to help the Rabbit to improve his running strategy for the next race to avoid falling asleep again.

Every hour, the Rabbit can either decide to sprint (run fast) or jog (run slow) for the next hour. If the Rabbit is currently *fresh*, he can decide to jog and will remain fresh. But if he decides to sprint for an hour, he has only an **0.5** chance of remaining fresh, and will end up *tired* otherwise. Once the Rabbit is tired, if he decides to slow down and starts to jog again for the next hour, there is an **0.5** chance that he will remain tired, and **0.5** chance that he will recover and end up *fresh* again. However, if the Rabbit is tired and continues sprinting for another hour, he will fall asleep at the end of the hour.

The Rabbit runs 0km per hour when he sleeps, 1 km per hour when he jogs and 2 km when he sprints. He therefore decides to assign a reward of **0**, **1** and **2** to, respectively, every hour that he sleeps, jogs and sprints. However, once he falls asleep, he does not finish the race. Said differently, being asleep counts as a terminal state: once the Rabbit reaches that state no further actions are possible and no further rewards are given.

To reason about different race strategies, you model this problem with the Markov decision process (MDP) depicted in Fig. 4. In particular, this MDP has the following three states:

- *Fresh* (F), where he can either jog (J) or sprint (S),
- *Tired* (T), where he can either jog (J) or sprint (S),
- *Asleep* (A), which is a terminal state.

We assume that the Rabbit did not party all night before the race and therefore always starts *fresh*, i.e., in state (F).

General advice for this secion: Wherever applicable, we recommend to do the calculations symbolically and in particular to keep the discount factor γ a parameter, so that intermediate results can be reused. However, in the end, you will be asked to provide numerical values.

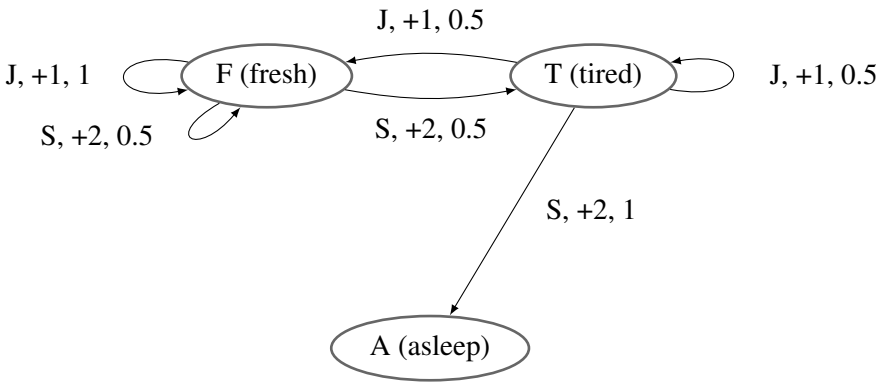


Figure 4: Rabbit's running strategy MDP. Each arrow is labeled with a letter, number (either +1, +2, or 0), and a number between 0 and 1. The letter indicates the action that corresponds to the arrow. The first number indicates the reward for transitioning between two states that also depends on the taken action. The second number indicates the probability of the transition for the given action.

(1 point) (i) How many different deterministic policies are there in total?

Solution: 4

(1 point) (ii) Consider two policies π_1, π_2 such that $\pi_1(F) = \pi_2(F) = J$ and $\pi_1(T) = J$ and $\pi_2(T) = S$. If he follows either of these two policies, starting in F, which states can he potentially visit?

◯ F and T √ **Only F** ◯ F and T and S ◯ Depends on the discount factor γ

(1 point) (iii) Is the value of the state F the same for the two policies, i.e., $V^{\pi_1}(F) = V^{\pi_2}(F)$ for every discount factor $\gamma \in (0, 1)$?

√ **Yes** ◯ No

Solution: Yes, the value is the same, since they never visit any other state than F and have the same policy on F.

(4 points) (iv) Consider now policy π_3 given by $\pi_3(\text{F}) = \text{S}$, $\pi_3(\text{T}) = \text{S}$ and $\gamma = 0.5$. What are the values of the states F, T and A, i.e., $V^{\pi_3}(F)$, $V^{\pi_3}(T)$ and $V^{\pi_3}(A)$?

- *(1 point)* $V^{\pi_3}(A) =$
- *(1 point)* $V^{\pi_3}(T) =$
- *(2 points)* $V^{\pi_3}(F) =$

Solution: For ease of notation, let us write

$$R_1 = r(T,S), \;\; R_2 = r(F,S), \;\; a = p(T|F,S), \;\; b = p(F|F,S).$$

$$V^{\pi_3}(A) = 0$$

$$V^{\pi_3}(T) = R_1 + \gamma V^{\pi_3}(A) \implies V^{\pi_3}(T) = 2.00$$

$$V^{\pi_3}(F) = R_2 + \gamma a V^{\pi_3}(T) + \gamma b V^{\pi_3}(F) \implies V^{\pi_3}(F) = \frac{R_1 + \gamma a V^{\pi_3}(T)}{1 - \gamma b} = \frac{2+\gamma}{1-0.5\gamma} = 3.33$$

(4 points) (v) Now consider policy π_4 given by π_4(F) = S, π_4(T) = J and $\gamma = 0.5$. What are the values of the states F and T, $V^{\pi_4}(F)$, $V^{\pi_4}(T)$?

- $V^{\pi_4}(F) =$
- $V^{\pi_4}(T) =$

Solution: For ease of notation, let us write

$$R_1 = r(F,S), \;\; R_2 = r(T,J), \;\; a = p(F|F,S), \;\; b = p(F|T,J),$$

Therefore,

$$V^{\pi_4}(F) = R_1 + \gamma a V^{\pi_4}(F) + \gamma(1-a)V^{\pi_4}(T) \implies V^{\pi_4}(I) = \frac{R_1 + \gamma(1-a)V^{\pi_4}(T)}{1 - a\gamma}$$
$$V^{\pi_4}(T) = R_2 + \gamma b V^{\pi_4}(F) + \gamma(1-b)V^{\pi_4}(T) \implies V^{\pi_4}(T) = \frac{R_2 + \gamma b V^{\pi_4}(F)}{1 - (1-b)\gamma}.$$

Then, it follows

$$V^{\pi_4}(F) = \frac{2 + 0.5\gamma V^{\pi_4}(T)}{1 - 0.5\gamma}$$
$$V^{\pi_4}(T) = \frac{1 + 0.5\gamma V^{\pi_4}(F)}{1 - 0.5\gamma},$$

and

$$V^{\pi_4}(F) = \frac{2 + 0.5\gamma \frac{1 + 0.5\gamma V^{\pi_4}(F)}{1-0.5\gamma}}{1 - 0.5\gamma} \implies V^{\pi_4}(F) = \frac{2 - 0.5\gamma}{1 - \gamma} = 3.5.$$

Finally, we have

$$V^{\pi_4}(T) = \frac{1 + 0.5\gamma V^{\pi_4}(F)}{1 - 0.5\gamma} \implies V^{\pi_4}(T) = \frac{1 + 0.5 * 3.5 * \gamma}{1 - 0.5\gamma} = 2.5.$$

(4 points) (vi) When sprinting, the Rabbit gets double reward: he accepts a high short-term reward at the risk of getting "closer" to fall asleep. For a fixed reward function, the optimality of the Rabbit's decisions then depends on the interaction between the transition probabilities and parameter γ.

Let us compare the two previous policies π_3 and π_4 for different values of γ. Recall that π_3 is given by $\pi_3(\text{F}) = \text{S}$, $\pi_3(\text{T}) = \text{S}$ and that π_4 is given by $\pi_4(\text{F}) = \text{S}$, $\pi_4(\text{T}) = \text{J}$. We say that π_4 is a better policy than π_3 if $V^{\pi_4}(x) \geq V^{\pi_3}(x)$ for all $x \in \{F, T, A\}$ and there is at least one state, $x' \in \{F, T, A\}$, such that $V^{\pi_4}(x') > V^{\pi_3}(x')$.

For each of the statements below, decide whether they are true or false.

(Hint: reuse calculations from previous steps.)

(a) π_4 is a better policy than π_3 for all $\gamma \in (0, 1)$. ◯ True √ **False**

Solution: When $\gamma = 0$, the value of a state is the value of the next reward, which is always $\geq$ for π_3 than for π_4: so π_3 is better than π_4 in that case. But in the limit where $\gamma \to 1$, the value of a state tends to be the sum of all the rewards onwards. For state T and policy π_3 this is always 2 (since you sprint, and then the episode terminates). But for π_4, the value of state T tends to infinity, since the run never ends. So π_4 is better than pi_3 when γ is close enough to 1. So, overall, neither π_3 nor π_4 is better for all values of γ.

(b) If the reward $r(T, S)$ of sprinting in T is modified such that $r(T, S) = 1$ and all the other transition probabilities and rewards are kept as in the original formulation of the MDP, then π_4 is better than π_3 for all $\gamma \in (0, 1)$. √ **True** ◯ False

Solution: If the reward for jogging and sprinting is the same in state T (tired), then it is better to jog. Otherwise the Rabbit falls asleep and doesn't get any future rewards anymore.

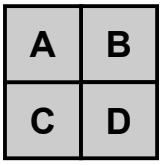
7. Reinforcement Learning (9 points)

(2 points) (i) Are the following methods for reinforcement learning *on-policy* or *off-policy*?

- Q-learning with epsilon-greedy exploration ◯ on-policy √ **off-policy**
- REINFORCE √ **on-policy** ◯ off-policy

> **Solution:** Q-learning is off-policy since it estimates the optimal Q-function with data from an arbitrary control policy. REINFORCE is a policy gradient method, and estimates the gradient with roll-outs from the current policy.

For the remaining questions we consider the four-state MDP shown below. The environment is episodic. The agent starts in state A and the episode ends if the agent reaches D.



In each state the agent has two actions, which deterministically move the agent to an adjacent state. For example, action "go right" ($\rightarrow$) in A moves the agent to B and action "go down" ($\downarrow$) in B moves the agent to D. Moving diagonally is *not* possible.

(4 points) (ii) Some policy produces the following transitions (not necessarily corresponding to fully completed episodes):

1. $(A, \rightarrow, +0, B)$,
2. $(B, \downarrow, +10, D)$,
3. $(A, \downarrow, +0, C)$,
4. $(C, \rightarrow, +20, D)$.
5. $(A, \downarrow, +0, C)$.

Your task is to compute Q-learning updates using the observations. Use the data in the order given. The Q-function is initialized with $Q_0(s, a) = 0$ for all $s \in \{A, B, C, D\}$ and the corresponding actions $a \in \{\rightarrow, \leftarrow, \uparrow, \downarrow\}$. Use learning rate $\alpha = 0.5$ and discount factor $\gamma = 1$. We use Q_t to refer to the Q-function after incorporating t transitions.

- $Q_5(A, \rightarrow) =$

> **Solution:** $Q_5(A, \rightarrow) = 0$ (Justification below.)

- $Q_5(A, \downarrow) =$

Solution: $Q_5(A, \downarrow) = 5$ (Justification below.)

Solution: For an observation quadruple (x, a, r, x'), the update is

$$Q_{i+1}(x, a) \leftarrow (1 - \alpha)Q_i(x, a) + \alpha(r + \gamma \max_{a'} Q_i(x', a'))$$

Note that data point 1. and 3. do not change the Q-function because all relevant quantities are zero. Hence, the non-trivial updates are

1. $Q_2(B, \downarrow) = 0.5(10) = 5$
2. $Q_4(C, \rightarrow) = 0.5(20) = 10$
3. $Q_5(A, \downarrow) = 0 + 0.5(0 + \gamma \max_{a'} Q_4(C, a')) = 5$

(3 points) (iii) For the same MDP, we use REINFORCE to optimize a policy π_θ with one-dimensional parameter $\theta \in \mathbb{R}$. In state A, the parameter is used to define a softmax probability for choosing action $\rightarrow$ or $\downarrow$ respectively. In state B the policy is defined to always choose $\downarrow$, and $\rightarrow$ in state C. Specifically,

- $\pi_\theta(\rightarrow | A) = \frac{\exp(\theta)}{\exp(\theta)+\exp(-\theta)}$
- $\pi_\theta(\downarrow | A) = \frac{\exp(-\theta)}{\exp(\theta)+\exp(-\theta)}$
- $\pi_\theta(\downarrow | B) = 1$
- $\pi_\theta(\rightarrow | C) = 1$

All other action probabilities are zero. The parameter is initialized with $\theta_0 = 0$. The agent then produces the following trajectory:

- $(A, \rightarrow, +0, B, \downarrow, +10, D)$

Compute the REINFORCE update θ_1 of θ obtained with this trajectory using a learning rate $\eta = 0.5$ and discount factor $\gamma = 1$.

- $\theta_1 =$

Solution: $\theta_1 = 5$.

For a trajectory $(x_1, a_1, r_1, x_2, a_2, r_2, x_3)$, we perform two gradient steps. Let $G_1 = r_1 + r_2$ and $G_2 = r_2$ be the rewards to go to state x_1 and x_2 respectively. Then the corresponding updates are

$$\theta \leftarrow \theta + \eta G_1 \nabla_\theta \log \pi_\theta(a_1|x_1)$$
$$\theta \leftarrow \theta + \eta G_2 \nabla_\theta \log \pi_\theta(a_2|x_2)$$

Note that the action choice in state B and C does not dependent on θ, therefore the second gradient step has no effect in our case.

So we get $G_1 = 10$ and

$$\theta_1 \overset{\leftarrow}{=} \theta_0 + \eta G_1 \nabla_\theta \log \left(\frac{1}{1+\exp(-2\theta)} \right) |_{\theta=\theta_0} \tag{3}$$

$$= 0 - 0.5 \cdot 10 \cdot \nabla_\theta \log \left(1+\exp(-2\theta)\right) |_{\theta=\theta_0} \tag{4}$$

$$= -5 \cdot \frac{-2\exp(-2\theta_0)}{1+\exp(-2\theta_0)} \tag{5}$$

$$= 10 \cdot \frac{1}{1+1} = 5 \tag{6}$$